Extended Data: Intercity Yb/Sr optical clock comparison over 1 350 km fiber link at 2.3×10^{-18}

xxxxx

1 Extended Data

Table 1: Per-segment data of the 16 segments analyzed in this work. Ratio deviations $y_i = R_i/R_1 - 1$ are relative to the first segment, with $R_1 = 1.207\,507\,039\,343\,337\,721\,0$. $u(y_i)$ is the statistical uncertainty of the segment ratio; "Fit" is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

Segment	n_{valid}	y_i (10^{-18})	$u(y_i)$ (10^{-18})	Fit
1	68 009	+0.00	4.245	est.
2	29 481	-3.36	3.017	lim.
3	42 891	+2.00	5.880	est.
4	64 280	-4.61	3.674	est.
5	19 532	-3.38	15.250	est.
6	52 140	-1.15	2.230	lim.
7	39 526	+5.31	3.476	lim.
8	43 565	+0.97	9.860	est.
9	57 938	-0.76	5.003	est.
10	35 824	+2.61	5.976	est.
11	40 346	+0.73	4.843	lim.
12	152 994	-3.09	4.019	est.
13	68 618	+1.97	6.254	est.
14	57 649	-0.25	4.028	est.
15	147 640	-0.77	1.809	lim.
16	65 983	-0.89	8.813	est.

Table 2: Leave-one-out sensitivity of the ratio combination. Each row excludes one of the 16 segments and recomputes the combination from the remaining segments. The center values are stable at the 10^{-18} level under any single exclusion.

Excluded segment	R_{wls}	χ^2_{red}	u_{wls} (10^{-18})
1	1.207 507 039 343 337 720 4	0.542	1.136
2	1.207 507 039 343 337 720 8	0.474	1.164
3	1.207 507 039 343 337 720 3	0.531	1.123
4	1.207 507 039 343 337 720 8	0.451	1.145
5	1.207 507 039 343 337 720 4	0.540	1.111
6	1.207 507 039 343 337 720 6	0.535	1.217
7	1.207 507 039 343 337 719 9	0.347	1.150
8	1.207 507 039 343 337 720 4	0.542	1.114
9	1.207 507 039 343 337 720 4	0.543	1.128
10	1.207 507 039 343 337 720 3	0.525	1.122
11	1.207 507 039 343 337 720 4	0.539	1.130
12	1.207 507 039 343 337 720 6	0.512	1.139
13	1.207 507 039 343 337 720 4	0.533	1.121
14	1.207 507 039 343 337 720 4	0.543	1.139
15	1.207 507 039 343 337 720 6	0.540	1.287
16	1.207 507 039 343 337 720 4	0.543	1.115

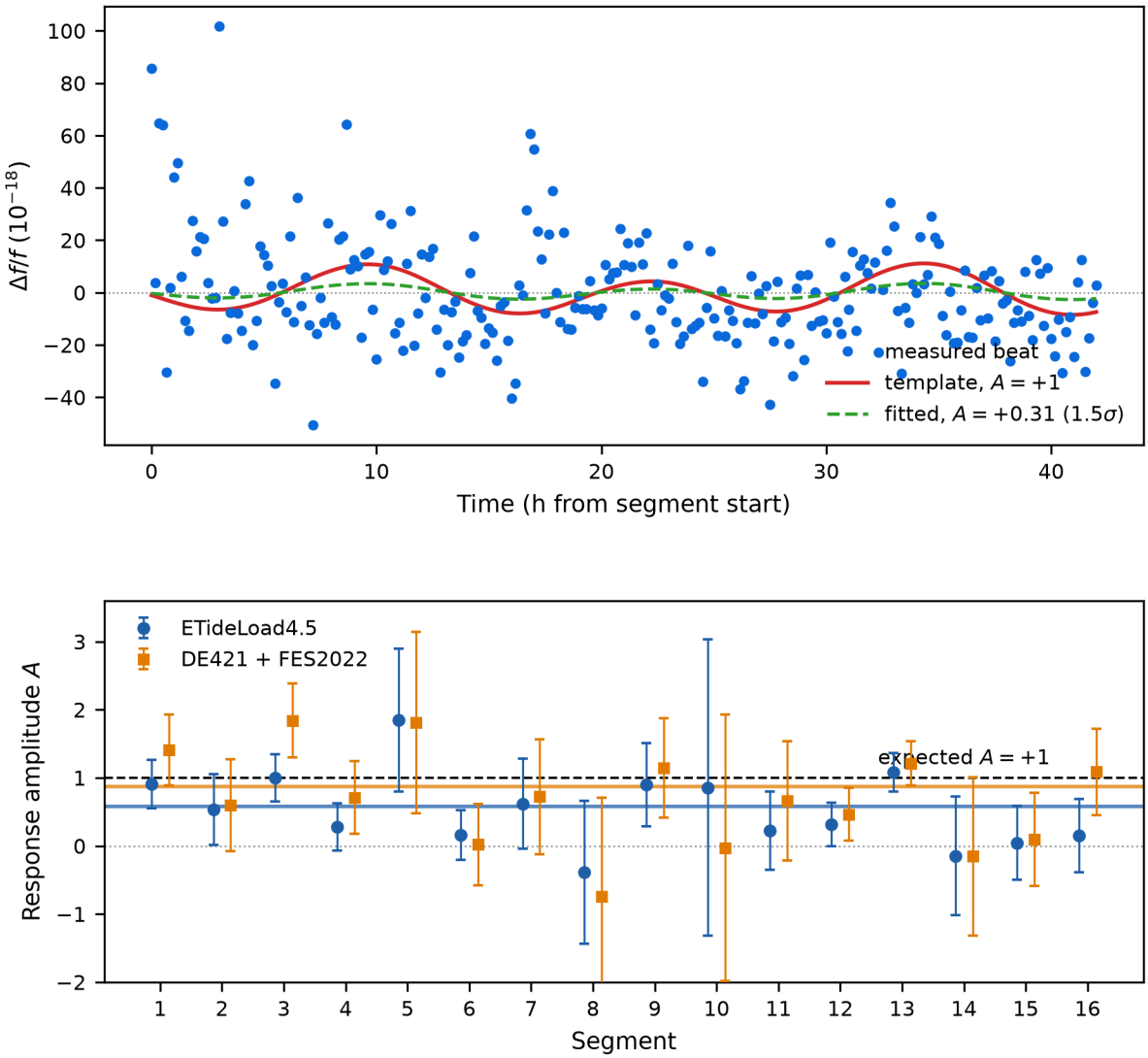


Figure 1: Within-segment tidal correlation analysis. **(a)** Single-segment example: the 1200-s demeaned beat waveform (dots) and the tidal template scaled at the expected response $A = +1$ and at the fitted amplitude, for the ETideLoad4.5 model. No single segment reaches a significant correlation on its own. **(b)** Per-segment response amplitudes A for the two tide models, over the 16 segments entering the ratio combination. Error bars are per-segment fit uncertainties from non-overlapping windows; the solid lines mark the precision-weighted combinations. In 12 of the 16 segments the correlation is positive for either model, and both precision-weighted amplitudes (+0.58 and +0.87) fall short of the expected response of +1 (dashed line), so the observed tidal coupling is smaller than any single template predicts.